

EDGE INTELLIGENCE

LAB - 4

25MML0043

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TASK 1: ARTIFICIAL NEURAL NETWORK USING DL

```
import tensorflow as tf
import numpy as np
# 1. Load MNIST dataset
data = np.load('mnist.npz')

x_train = data["x_train"]
y_train = data["y_train"]
x_test = data["x_test"]
y_test = data["y_test"]

# 2. Normalize the pixel values (0-255 → 0-1)
x_train = x_train / 255.0
x_test = x_test / 255.0

model = tf.keras.Sequential([
    tf.keras.layers.Flatten(input_shape=(28, 28)),
    tf.keras.layers.Dense(128, activation='relu'),
    tf.keras.layers.Dense(10, activation='softmax')
])

model.compile(optimizer='adam',
              loss='sparse_categorical_crossentropy',
              metrics=['accuracy'])

model.fit(x_train, y_train, epochs=5)
```

```
Epoch 1/5
1875/1875 — 8s 4ms/step - accuracy: 0.9247 - loss: 0.2617
Epoch 2/5
1875/1875 — 7s 4ms/step - accuracy: 0.9658 - loss: 0.1150
Epoch 3/5
1875/1875 — 7s 4ms/step - accuracy: 0.9765 - loss: 0.0785
Epoch 4/5
1875/1875 — 6s 3ms/step - accuracy: 0.9822 - loss: 0.0581
Epoch 5/5
1875/1875 — 6s 3ms/step - accuracy: 0.9864 - loss: 0.0437
```

```
<keras.src.callbacks.history.History at 0x22b0695e440>
```

```
#Evaluate model
test_loss, test_acc = model.evaluate(x_test, y_test, verbose=0)

print(f'\nTest Accuracy: {test_acc*100:.4f}')
```

```
print(f'Test Loss: {test_loss*100:.4f}')
```

```
#Saving model
import pickle

# Save the model
with open("mnist_model.pkl", "wb") as f:
    pickle.dump(model, f)

# To load it back later:
# with open("mnist_model.pkl", "rb") as f:
#     loaded_model = pickle.load(f)
```

```
... Loading model from mnist_model.pkl...
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/mnist.npz
11490434/11490434 ————— 2s 0us/step

Evaluating loaded model performance...
Restored model accuracy: 92.08%

Testing on first 5 images...
1/1 ————— 0s 82ms/step
Predicted: [7 2 1 0 4]
Actual: [7 2 1 0 4]
```

```
> import os

file_path = "mnist_model.pkl"
# getsize returns bytes, divide by 1024 for KB
size_kb = os.path.getsize(file_path) / 1024

print(f"File Size: {size_kb:.2f} KB")
```

```
[4] ✓ 0.0s
```

```
... File Size: 417.60 KB
```

```
model.summary()
```

[5] ✓ 0.0s

Model: "sequential"

Layer (type)	Output Shape	Param #
flatten (Flatten)	(None, 784)	0
dense (Dense)	(None, 128)	100,480
dense_1 (Dense)	(None, 10)	1,290

Total params: 101,772 (397.55 KB)

Trainable params: 101,770 (397.54 KB)

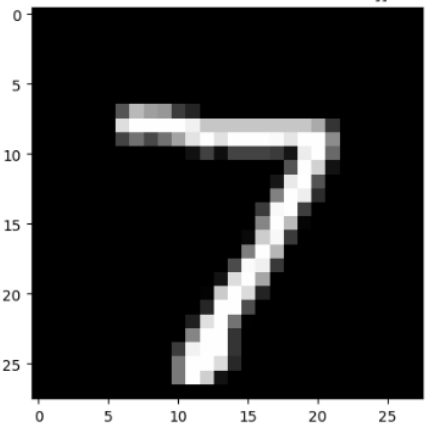
Non-trainable params: 0 (0.00 B)

Optimizer params: 2 (12.00 B)

```
import matplotlib.pyplot as plt
plt.imshow(x_test[0], cmap='gray')
plt.title("predicted:" + str(pred))
```

✓ 0.5s

Text(0.5, 1.0, 'predicted:[[2.75356637e-04 1.69666248e-06 2.37161395e-04 1.73411972e-03\n 2.40992995e-05 8.68187635e-05 9.65907248e-07 9.94599342e-01\n 1.27018560e-04 2.91345967e-03]\n [3.54252290e-03 1.65653692e-04 8.88520598e-01 2.14903187e-02\n 3.64339212e-06 1.29046664e-02 6.48955032e-02 1.60882678e-07\n 8.47490225e-03 2.03188733e-06]\n [1.30446148e-04 9.68027294e-01 1.10185165e-02 3.98947159e-03\n 7.58480746e-04 1.76262611e-03 3.21953674e-03 3.53753963e-03\n 6.44446723e-03 1.11162977e-03]\n [9.98167276e-01 4.62121896e-09 1.48636289e-04 4.60250521e-05\n 2.46783685e-07 1.08933903e-03 3.38261423e-04 6.30978029e-05\n 9.91277120e-05 4.79449554e-05]\n [1.96933770e-03 2.54092865e-05 9.73366201e-03 5.40680077e-04\n 8.95198882e-01 1.14605890e-03 5.85654425e-03 7.27365073e-03\n 9.97465756e-03 6.82810321e-02]]')



TASK 2: EDGE IMPLUSE

STEPS:

1. Create an account on **Edge Impulse**.
2. Login to the Edge Impulse platform.
3. Create a **new project**.
4. Click **Data Acquisition**.
5. Select **Connect to device**.
6. Choose **Use mobile phone** and scan the QR code.
7. Connect the mobile phone as a camera.
8. Capture sample images.
9. Collect images for **training** and **testing**.
10. Rename images with meaningful names.
11. Label all images correctly.

Training -8

The screenshot displays the Edge Impulse Studio web interface. The top navigation bar includes links for Dataset, Data explorer, Data sources, Synthetic data, AI labeling, and CSV Wizard. The main content area shows a dataset with 19 items and a train/test split of 67% / 33%. A table lists the dataset items, including sample names, labels, and added dates. A sidebar on the left contains navigation links for Dashboard, Devices, Data acquisition, Experiments, EON Tuner, Impulse design, Versioning, and Getting started. A 'Collect data' section on the right prompts the user to connect a device. A 'RAW DATA' section at the bottom right shows a list of raw data samples.

SAMPLE NAME	LABEL	ADDED
Fan	Mouse	Yesterday, 17:17:51
unknown.6ebbvt4	Mouse	Yesterday, 17:14:02
unknown.6ebbqfv	Mouse	Yesterday, 17:13:23
unknown.6ebb9qs	Light	Yesterday, 17:12:44
unknown.6ebb95n1	Light	Yesterday, 17:12:29
unknown.6ebb8b28	Light	Yesterday, 17:12:02
unknown.6ebb7oo0	Fan	Yesterday, 17:11:43
Fan	Fan	Yesterday, 17:11:04
unknown.6ebb5j1	Light	Yesterday, 17:10:32
unknown.6ebb1jmg	Light	Yesterday, 17:08:21
unknown.6ebau5m1	Fan	Yesterday, 17:06:29

Testing-6

The screenshot displays the Edge Impulse web interface. On the left is a sidebar with navigation links: Dashboard, Devices, Data acquisition, Experiments, EON Tuner, Impulse design (with a dropdown menu containing Create impulse, Live classification, Model testing, and Deployment), Versioning, GETTING STARTED, Documentation, and Upgrade Plan. The main content area has a top navigation bar with tabs: Dataset (selected), Data explorer, Data sources, Synthetic data, AI labeling, and CSV Wizard. Below this, a summary section shows 'DATA COLLECTED: 19 Items' and 'TRAIN / TEST SPLIT: 67% / 33%'. A 'Collect data' section on the right prompts to 'Connect a device to start building your dataset.' The central 'Dataset' section features a table with columns 'SAMPLE NAME', 'LABEL', and 'ADDED'. It lists six samples, all labeled 'unknown', with their respective IDs and timestamps. A 'RAW DATA' section on the right is currently empty, showing a 'Click on a sample to load...' message.

EDGE IMPULSE

Ganesh_AWH / Ganesh_AWH-project-1 PERSONAL Target: Cortex-M4F 80MHz

Dataset Data explorer Data sources Synthetic data AI labeling CSV Wizard

DATA COLLECTED: 19 Items

TRAIN / TEST SPLIT: 67% / 33%

Collect data

Connect a device to start building your dataset.

RAW DATA

Click on a sample to load...

Dataset

Training (10) Test (9)

SAMPLE NAME	LABEL	ADDED
unknown.6ebbj4la	Mouse	Yesterday, 17:27:03
unknown.6ebbj38q	unknown	Yesterday, 17:17:53
unknown.6ebbg46o		Yesterday, 17:16:17
unknown.6ebbdnso	unknown	Yesterday, 17:14:59
unknown.6ebbc1c5	unknown	Yesterday, 17:14:24
unknown.6ebba9ck	unknown	Yesterday, 17:13:06

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